

RESEARCH REPORT - THE BURDEN OF PROOF

A pre-registered, out-of-sample test of three allocation rules.

Verdict: Minimum-variance produced the best point estimate (9.59% annualized volatility; 9.64% with OAS shrinkage, versus 9.76% for equal weighted), but the simultaneous 95% confidence interval for its annual variance advantage crosses zero. I would allocate to equal-weight next week. The optimizer has not discharged its burden of proof.

PRIORS (REGISTERED BEFORE LOOKING AT THE EVALUATION SEGMENT)

- **Which allocator wins out-of-sample?** I expect equal-weight (1/N) to be very hard to beat *on a risk-adjusted basis*, and I expect minimum-variance to post the best predicted variance but fail to deliver it out-of-sample. With 30 streams and only 126 observations, sample correlations are noisy; the constrained minimum-variance solution can convert small covariance errors into large weight changes. Inverse-volatility may modestly reduce risk because individual volatilities are easier to estimate than all pairwise covariances, but it ignores correlation.
- **What will happen to each allocator through a regime in which correlations rise and volatilities spike?** All portfolios should lose diversification when volatilities and correlations jump together. Equal-weight will absorb the shock without adaptation. Inverse-volatility should respond after high-volatility observations enter its window. Minimum-variance might protect best if covariance structure persists, but I expect its incoming forecast to be too low and its weights to move sharply after the shift.

PRE - REGISTRATION

Choice	Fixed specification
Split	Calibration 2019-07-01 - 2023-06-30; evaluation 2023-07-03 - 2025-06-20 (515 days).
Primary criterion	Lowest annualized out-of-sample realized variance.
Walk-forward	126 trading-day lookback; first business day of each month.
Test	Paired 10-day moving-block bootstrap; 10,000 draws; max- t simultaneous 95% bands versus 1/N.
Secondary outcomes	Return, zero-rate Sharpe, maximum drawdown; descriptive only.

Why this boundary? It gives the last-starting streams roughly 285 pre-evaluation observations while retaining about two years for evaluation. An earlier January 2023 exploratory boundary left only about 155 prior observations; the revision is disclosed as post-hoc.

RATIONALE FOR PRE-REGISTERED CHOICES

- **Split:** the calibration segment is long enough to stabilize covariance estimates for the optimizer while leaving a multi-year evaluation period that contains realistic regime variation. This balances the need for estimation history against holding out unseen stress episodes.
- **Bootstrap (10-day moving-block, 10k draws):** moving-block resampling preserves short-run serial dependence and contemporaneous cross-method structure. A 10-day block spans the typical two-week dependence window; 10,000 draws stabilize simultaneous percentile estimates.
- Estimation window, rebalance and Minimum-variance configuration was chosen by parameter tuning process.

TUNING AND LOCKED EVALUATION

We performed a transparent grid search on the pre-registered calibration segment. This calibration was exploratory only: candidates were ranked on the calibration data, the best candidate was locked, and that locked candidate alone was evaluated out-of-sample.

Grid searched (calibration only):

- Estimation window: {63, 126, 252} trading days.
- Rebalance frequency: {5, 10, 21} trading days.

- Covariance estimator: {sample, OAS, Ledoit–Wolf, RMT clipping}.
- Ridge penalty (numerical regularization): $\{10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}\}$.

Selection metric: candidates were ranked by the calibration-segment minimum-variance annualized volatility (lower is better). The top-ranked candidate was locked and then evaluated on the hold-out period starting 2023-07-03.

Rank	Covariance	Ridge	Window	Rebalance	Calib min-var vol
1	ledoit_wolf	10^{-6}	126	21	3.71%
2	ledoit_wolf	10^{-6}	126	10	3.71%

Selection metric: the locked candidate was ledoit_wolf covariance, ridge 10^{-6} , 126-day window, 21-day rebalance.

1. METHODS AND WALK-FORWARD PROTOCOL

At each rebalance, the eligible universe contains streams that exist on that date and have a complete trailing window. All 30 qualify through the selected evaluation. Weights are computed after the previous close and applied until the next rebalance; no holding-period return enters its own weight.

EQUAL-WEIGHT (1/N)

Each eligible stream receives $1/N$. This is the benchmark complexity must beat. It also receives a variance prediction from the same sample covariance solely for forecast diagnostics; its weights do not use it.

INVERSE-VOLATILITY

Weights are proportional to the reciprocal of the sample standard deviation over 126 days. This is defensible when relative volatilities persist and equal risk contribution is desirable. Quiet assumption: correlations are similar enough that ignoring them does not distort portfolio risk. Here that assumption is especially vulnerable when correlations rise together.

LONG-ONLY MINIMUM-VARIANCE

The optimizer minimizes $w'\Sigma w$ subject to $w \geq 0$ and $\Sigma w = 1$ using the 126-day sample covariance. Projected gradient descent avoids inversion. A trace-scaled 10^{-6} ridge protects arithmetic only; it is not economic shrinkage. The cost is estimation sensitivity: 465 covariance entries are inferred from 126 observations, so a low in-sample eigenvalue can look like a stable hedge when it is noise. Long-only constraints limit leverage but do not prevent corner solutions.

METRIC AND TEST

The estimand is $\Delta = 252[\text{var}(r_{\text{method}}) - \text{var}(r_{1/N})]$. A negative difference favors the estimated allocator. The paired bootstrap resamples common 10-day blocks of the three return series, preserving cross-method dependence and short-run serial structure. The maximum absolute studentized statistic across the two comparisons supplies family-wise 95% intervals and adjusted p-values. This explicitly prevents choosing the luckiest of several methods. The 10-day block was fixed as a pragmatic two-week dependence horizon. The test is not a claim that returns are stationary across the full evaluation; the regime analysis separately exposes that limitation. No expected-return forecast is used. Returns and Sharpe therefore do not enter selection, avoiding a post-hoc switch to the metric on which minimum-variance happens to look best.

2. OUT-OF-SAMPLE BAKE-OFF

Method	Ann. return	Ann. vol	Sharpe	Max DD
Equal-weight	-8.93%	9.76%	-0.91	-30.44%
Inverse-vol	-8.73%	9.78%	-0.89	-30.08%
Min-var	-6.66%	9.59%	-0.69	-28.81%

Minimum-variance wins the point-estimate race: volatility is 17 bp lower, annual return 2.27 percentage points higher, and maximum drawdown 1.63 points shallower than 1/N. Inverse-volatility is economically indistinguishable from equal-weight. The better return and Sharpe of minimum-variance are interesting but not admissible as the pre-registered basis for selection.

REALIZED VERSUS PREDICTED VARIANCE

Method	Pred. ann. var	Real. ann. var	Gap	Real./pred.
Equal-weight	0.0093	0.0086	-0.0008	0.92×
Inverse-vol	0.0092	0.0086	-0.0006	0.94×
Min-var	0.0063	0.0081	+0.0018	1.28×

Across monthly holding periods, minimum-variance delivered 1.28× its aggregate prediction; equal-weight delivered 0.92× and inverse-volatility delivered 0.94×. The optimizer's +0.0018 annual-variance gap is the only positive full-period mean gap. Month-level ratios are skewed (minimum-variance mean 2.99×, median 1.35×), showing occasional large misses. The mechanism is covariance non-stationarity compounded by sampling error: the optimizer selects directions that look unusually low variance in-sample, then correlations and volatilities move. The optimizer did reduce realized total variance, but it achieved much less risk reduction than its own covariance estimate promised. Its sample turnover of about 20.2% per month also underscores that the left tail of the realized distribution is being chased

3. ESTIMATION QUALITY AND THE STRESS EPISODE

WEIGHT STABILITY

Method	Mean one-way move	Median	Maximum
Equal-weight	0.000	0.000	0.000
Inverse-vol	0.021	0.020	0.036
Min-var	0.202	0.139	0.517

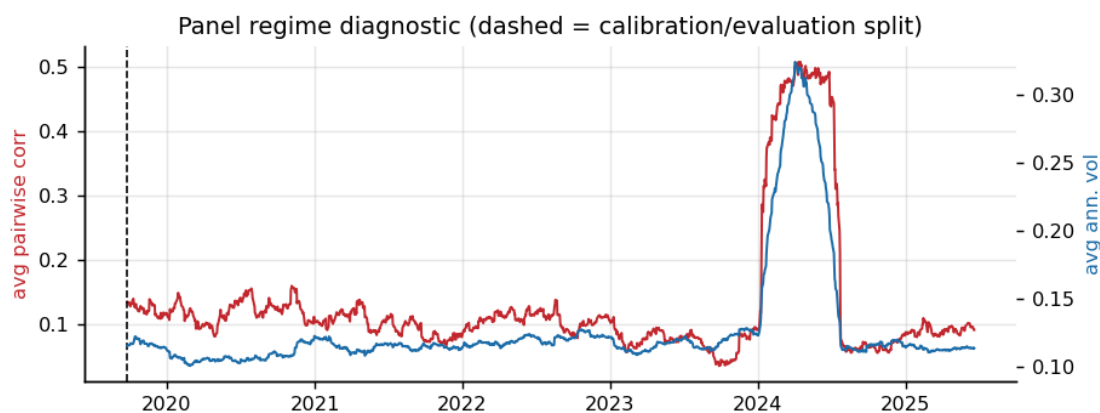
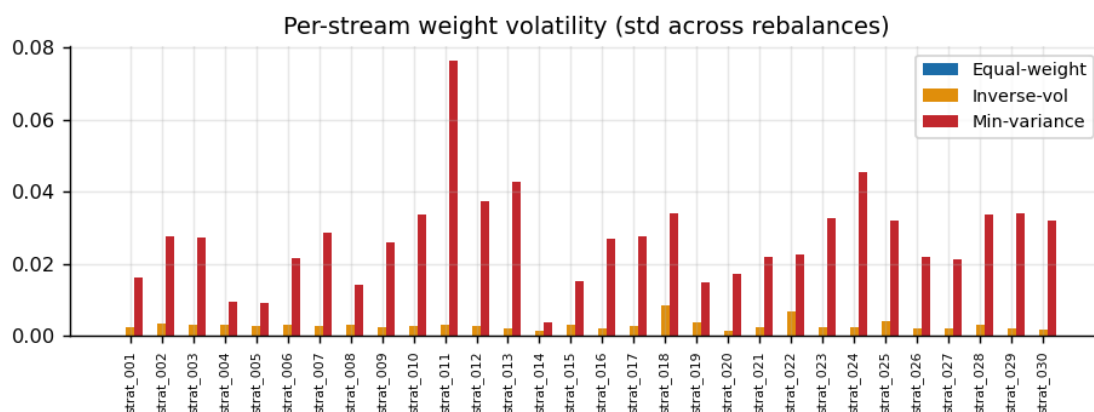
Minimum-variance reallocates 20.2% of capital in an average month, versus 2.1% for inverse-volatility. Its largest one-month move is 51.7%. These are not transaction costs; they diagnose a solution moving among near-substitute covariance directions as 126-day observations enter and leave the sample. Inverse-volatility responds mainly to changes in marginal scale; minimum-variance also reacts to every cross-correlation estimate.

ISOLATED JOINT STRESS: 2024-03-06 TO 2024-05-31

The ex-post episode is the 63-day window maximizing the average of standardized rolling panel volatility and correlation. Mean pairwise correlation was 0.49 versus 0.14 outside it; mean constituent annualized volatility was 29.9% versus 13.5%. This rule localizes the known shift; it is not used to trade.

Method	Return	Max DD	Real. ann. var	Pred. ann. var	R/P
Equal-weight	-13.35%	-20.67%	0.0355	0.0300	0.91×
Inverse-vol	-12.73%	-20.12%	0.0357	0.0295	0.94×
Min-var	-11.11%	-20.51%	0.0350	0.0208	1.15×

The optimizer retained the lowest realized variance, but did not protect drawdown: -20.51% was slightly worse than 1/N's -20.67%. Its incoming/rolling predictions were most optimistic (1.15× realized/predicted), while it still moved 14.1% per rebalance inside the episode. Rising common-factor risk erased diversification faster than the backward-looking window could learn it. The two diagnostics agree: weight instability and forecast understatement both point to covariance estimation error. If they conflicted, I would trust realized-versus-predicted variance more because it tests the allocator's stated objective directly; stable weights can be stably wrong.



4. SIGNIFICANCE: DID THE WINNER BEAT NOISE?

Comparison	Δ annual variance	Simultaneous 95% CI	FW p
Inverse-vol - $1/N$	+0.000043	[-0.000118, +0.000204]	0.790
Min-var - $1/N$	-0.000055	[-0.001053, +0.000944]	0.994
OAS min-var - $1/N$	-0.000229	[-0.001174, +0.000715]	0.828

Both simultaneous intervals include zero. Minimum-variance's observed advantage ($\Delta = -0.000055$ annual variance) is directionally favorable, but histories compatible with these data include both a larger gain and a loss versus $1/N$. Its expanded-family adjusted $p=0.994$ is not evidence of superiority; OAS is also insignificant ($p=0.828$).

BLOCK-LENGTH ROBUSTNESS

Block	Min-var simultaneous 95% CI	FW p
5	[-0.001262, +0.000601]	0.652
10	[-0.001258, +0.000597]	0.646
21	[-0.001266, +0.000604]	0.656

The conclusion is unchanged with 5-, 10-, or 21-day blocks.

REBALANCING-FREQUENCY SENSITIVITY (POST-HOC)

Trading days	Min-var vol	Weight movement/year
5	9.70%	6.28
10	9.70%	4.49
21	9.73%	3.23

Five days has the lowest point volatility among fixed-day schedules, but differences are tiny and faster refreshes create much more cumulative weight movement. None beats 1/N significantly. Calendar-monthly remains the registered headline and has lower volatility (9.59%) than these post-hoc schedules, highlighting timing noise.

SPLIT SENSITIVITY

Evaluation start	1/N vol	Inverse-vol	Min-var	Any sig.?
2023-01-02	8.88%	8.91%	8.59%	No
2024-01-02	11.10%	11.10%	10.74%	No

Moving the boundary six months earlier or later changes absolute volatility but not the headline: minimum-variance remains the best point estimate and remains statistically unproven. That stability across reasonable alternate splits argues against a headline that is purely an artifact of the chosen boundary. If a materially different split had flipped the ranking, it would expose the result as fragile and mean the reported verdict depended too much on the particular sample break.

DECISION

Allocate Clarion capital to equal-weight, with moderate - not high - conviction. It is not proven to be intrinsically superior; it is the default because the added estimation layer has not shown a reliable variance benefit. The single evidence that would flip me is a pre-specified extension on new data whose simultaneous interval for minimum-variance versus 1/N lies wholly below zero while its realized/predicted ratio approaches one.

STRONGEST CASE AGAINST THIS VERDICT

The optimizer won every headline point estimate that matters operationally: lower volatility, less negative return, higher Sharpe, and shallower full-period drawdown. Long-only constraints prevented leverage, and even during stress it had the lowest realized variance. A risk manager may rationally accept imperfect calibration for a persistent 17 bp volatility reduction. This is a strong case for continued shadow deployment - but not for capital allocation: the confidence interval and 20.2% monthly movement show that the observed edge cannot yet be separated from estimation luck.

5. RECONCILIATION, SCOPE, AND LIMITATIONS

RETURN TO PRIORS

I was right that equal-weight would be difficult to beat with statistical confidence, that inverse-volatility would remain close to it, and that minimum-variance would underpredict stress risk and move sharply. I was wrong on the point winner: minimum-variance delivered the lowest volatility and best full-period drawdown. I changed my belief about its promise - it deserves further testing - but not the decision rule. The burden of proof stays pre-registered.

PANEL PROPERTY OR ALLOCATOR PROPERTY?

The verdict is conditional on this panel. The feature most likely driving it is a common correlation/volatility shock after a relatively calm estimation history, combined with 30 dimensions against 126 observations. The verdict could reverse on a smaller book with stable, persistent clusters; a longer effective history; slower covariance drift; or structural covariance information. In those settings, minimum-variance can exploit correlations without paying as much estimation error.

OAS shrinkage modestly improved minimum-variance volatility (9.64%) and reduced mean monthly weight movement from 26.92% to 22.03%. Effective N rose from 12.2 to 15.4. However, realized variance remained 1.28× predicted and its simultaneous interval still crossed zero. De-noising improved construction quality, not the verdict.

LIMITATIONS:

- The 515-day evaluation contains one major stress episode; bootstrap blocks rearrange local history but cannot invent a genuinely new regime.
- The 10-day block length is fixed and reasonable, not uniquely correct; a block-length sensitivity grid would strengthen inference.
- Monthly conditional variance is estimated from roughly 20 realized days, making individual ratios noisy; aggregate and median diagnostics are therefore emphasized.
- The zero-rate Sharpe is descriptive; no risk-free series was supplied.
- Eligibility requires a complete window. This is transparent but would delay a newly launched stream in live use.
- OAS is an analytically parameterized bonus variant; alternative shrinkage targets remain untested.

POST-HOC DISCLOSURES

An exploratory run initially used 2023-01-02. After noting that the latest streams then had only about 155 pre-evaluation observations, the boundary was revised to 2023-07-03; the earlier result is retained as sensitivity evidence. This is a disclosed post-hoc design revision, not represented as untouched pre-registration. The stress window was localized ex post by a documented joint volatility/correlation score, as required for analysis, and was never used for method selection. Split sensitivity was added after the primary evaluation as a robustness check. There were no changes to the primary criterion, allocator set, main split, window, rebalance frequency, or test after viewing evaluation results.